
7.2 How can materials be manipulated and joined in different ways in a workshop environment when making final prototypes?

- a. The use of specialist techniques, hand tools and equipment used to shape, fabricate, construct and assemble high quality prototypes, with exemplification of the following processes:
- i. wastage, such as:
 - o paper and boards, e.g. cutting and punching
 - o timber, e.g. sawing, drilling and turning
 - o metals, e.g. sawing, drilling, sheering and turning
 - o polymers, e.g. sawing and drilling
 - o fibres and fabrics, e.g. cutting and shearing
 - o design engineering, e.g. etching*.
 - ii. addition, such as:
 - o paper and boards, e.g. adhesion and laminating
 - o timber, e.g. adhesion, joining and laminating
 - o metals, e.g. adhesion, welding/brazing and riveting
 - o polymers, e.g. adhesion and heat welding
 - o fibres and fabrics, e.g. sewing, bonding and laminating
 - o design engineering, e.g. soldering*.
 - iii. deforming and reforming, such as:
 - o paper and boards, e.g. perforating and folding
 - o timber, e.g. steaming and pressing
 - o metals, e.g. pressing, bending and casting
 - o polymers, e.g. moulding, vacuum forming and line bending
 - o fibres and fabrics, e.g. heat treatments, pleating and gathering
 - o design engineering, e.g. moulding*.

** and other processes appropriate to the materials or components being used in design engineering.*
